

The CRR interval law

A lemma-by-lemma verification, and a unification of the two competing variants

The Coherence–Rupture–Regeneration corpus contains **two incompatible predictions for the same quantity**. Its May 2026 validation supplement uses $CV = \Omega/2$ across twenty systems and calls the resulting class ratio of 2 a topological invariant. Its August 2026 papers use $CV = \sqrt{\Omega}$, a ratio of $\sqrt{2}$, and name a ratio of 2 as a *falsifier*. Both cannot be right as stated. This note establishes that both are correct in their own regime, that they are the two limits of a single expression, and that the expression is a standard result in renewal theory.

Verification protocol.

Every mathematical claim below is stated as a lemma, matched to a named theorem in the published literature, and checked numerically against its analytic value. Thirteen lemmas, thirty-eight individual checks. **Thirty-seven passed on the first run. One failed.** The failure was in this note's own statement of Hurwitz's theorem, not in the theorem; it is documented in §7 rather than quietly repaired, because a verification document that reports no errors has usually not verified anything. After correction, thirty-eight of thirty-eight pass, and the central result (Theorem 1) reproduces in simulation to within 0.05%.

1. Notation, and the two variants

Both variants agree on the CRR skeleton. Coherence accumulates on a statistical manifold; the regime holds $a = C^* = 1/\Omega$ resolvable cells; rupture falls at $C \cdot \Omega = 1$. They disagree about *where the randomness lives*, and that is the whole of the disagreement.

	V1 wall-noise	V2 quantal
accumulation	deterministic	stochastic (Poisson quanta)
threshold	jittered, Bernoulli($1/2$)	fixed at $1/\Omega$
interval law	$CV = \Omega/2$	$CV = \sqrt{\Omega}$
distribution	near-deterministic, symmetric	Gamma, shape $1/\Omega$
skewness	≈ 0	$2\sqrt{\Omega}$
Z_2 ($\Omega=1/\pi$)	0.1592	0.5642
$SO(2)$ ($\Omega=1/2\pi$)	0.0796	0.3989
class ratio	2	$\sqrt{2}$
where used	20-system supplement, May 2026	mortal computation & dossier, Aug 2026

Neither is a mistake. They are different physical models that happen to share a threshold condition, and the corpus moved from one to the other between May and August without noticing that a single expression contains both.

2. The lemma register

Each lemma is stated, sourced to the standard literature, and verified numerically. “dev” is the largest relative deviation of simulation from the analytic value across all parameter settings tested.

L1. Fisher–Rao extent of the Bernoulli manifold is π

The Fisher information of Bernoulli(p) is $I(p) = 1/[p(1-p)]$, so $ds = dp/\sqrt{p(1-p)}$. Substituting $p = \sin^2\theta$ gives $ds = 2 d\theta$ on $\theta \in [0, \pi/2]$, hence total extent π . Verified by quadrature to 10^{-8} and in closed form to machine precision. *Convention warning:* under the amplitude embedding $p \rightarrow 2\sqrt{p}$ with Fubini–Study normalisation the extent is $\pi/2$. The corpus uses π ; results scale accordingly. This is the Chentsov constant [1] and it must be fixed once and stated.

L2. A sum of k i.i.d. Exp(r) is Gamma(k, r), with $CV = 1/\sqrt{k}$ and skew = $2/\sqrt{k}$

Standard. Verified at $k = \pi, 2\pi, 20$ over 2×10^6 samples each; dev < 0.9% on skew, < 0.1% on CV. Setting $k = 1/\Omega$ gives the quantal law directly.

L3. Wald's equation and the Blackwell–Girshick equation

For $S = \sum_{i=1..N} X_i$ with N independent of the i.i.d. X_i , $E[S] = E[N]E[X]$ (Wald 1944) and $\text{Var}(S) = E[N]\text{Var}(X) + \text{Var}(N)E[X]^2$ (Blackwell & Girshick 1946) [2, 3]. These are the two identities Theorem 1 rests on.

L4. For a renewal process, the asymptotic Fano factor equals the squared CV of the interval

$F = \lim_{t \rightarrow \infty} \text{Var}[N(t)]/E[N(t)] = \text{CV}_X^2$. Verified by simulating renewal processes with $\text{CV}_X = 0.5, 1.0, 1.5$ and counting over long windows; dev < 5%. **This lemma has a real limitation and it must be carried forward:** the identity is asymptotic and holds for *renewal* processes. For non-renewal (correlated) event streams $F \neq \text{CV}_X^2$ in general [4, 5]. Any system with serially correlated evidence arrival falls outside Theorem 1 as stated.

L5. Bernoulli(1/2) has standard deviation 1/2

$\text{Var} = p(1-p) = 1/4$. Exact. This supplies the $\sigma_{\text{th}} = 1/2$ that the 20-system supplement admits it cannot derive from first principles.

L6. $B(C) = e^{C/\Omega}(C^* - C)$ is maximised at $C = C^* - \Omega$

$dB/dC = e^{C/\Omega}[(C^* - C)/\Omega - 1]$, zero at $C = C^* - \Omega$. Verified by 4×10^6 -point evaluation at three values of Ω ; agreement to 10^{-5} .

L7. Hurwitz's theorem and the three-gap theorem

Hurwitz (1891): for every irrational ξ there are infinitely many coprime p, q with $|\xi - p/q| < 1/(\sqrt{5} q^2)$, and $1/\sqrt{5}$ is optimal — it fails for any smaller constant when ξ is equivalent to the golden ratio [6]. Equivalently $\liminf_{q \rightarrow \infty} q \cdot \|q\xi\| \leq 1/\sqrt{5}$. Three-gap theorem (Steinhaus; proved by Sós, Surányi, Swierczkowski): the points $\{n\alpha\}$ for $n = 1..N$ partition the circle into at most three distinct gap lengths, the largest being the sum of the other two [7]. Both verified; see §7 for an error caught here.

L8. The inverse Gaussian (Brownian passage time) has skew = $3 \cdot \text{CV}$

For $IG(\mu, \lambda)$: $\text{CV} = \sqrt{(\mu/\lambda)}$ and skewness = $3\sqrt{(\mu/\lambda)}$. Verified at three values of λ ; dev < 1.2%. This is the seismological BPT model [8] and it is the third distinguishable variant.

L9. Summing adjacent terms scales the CV by $\sqrt{[(1+p)/2]}$

For X, Y with common mean μ , sd σ and correlation ρ , $\text{Var}(X+Y) = 2\sigma^2(1+\rho)$, so $\text{CV}(X+Y) = \text{CV}(X) \cdot \sqrt{[(1+\rho)/2]}$. At $\rho = 0$ this is $1/\sqrt{2}$. Verified at three correlations; dev < 1%.

3. Theorem 1: the unified interval law

Theorem 1. Let coherence accumulate as a renewal process of unit cells with mean rate μ , interval CV_X , and asymptotic Fano factor $F = CV_X^2$. Let the rupture threshold be $a + \xi$, where $a = 1/\Omega$ and ξ is independent of the accumulation with mean 0 and variance σ_{th}^2 . Then the rupture interval T satisfies

$$CV(T)^2 = F \cdot \Omega + \sigma_{th}^2 \cdot \Omega^2$$

Proof. By Wald (L3), $E[T] = E[a + \xi]/\mu = a/\mu$. By Blackwell–Girshick (L3), with X the per-cell waiting time of mean $1/\mu$ and variance σ_X^2 ,

$$Var(T) = E[a + \xi] \cdot \sigma_X^2 + Var(a + \xi) \cdot (1/\mu)^2 = a\sigma_X^2 + \sigma_{th}^2/\mu^2.$$

Dividing by $E[T]^2 = a^2/\mu^2$ and using $CV_X^2 = \mu^2\sigma_X^2 = F$ (L4):

$$CV(T)^2 = [a \cdot CV_X^2 + \sigma_{th}^2]/a^2 = F/a + \sigma_{th}^2/a^2 = F\Omega + \sigma_{th}^2\Omega^2. \blacksquare$$

Numerical verification of Theorem 1

Threshold $\theta \sim N(a, \sigma_{th})$ continuous; $T | \theta \sim \text{Gamma}(\theta/CV_X^2, CV_X^2)$; $a = \pi$; 4×10^6 samples per row.

F	σ_{th}	CV predicted	CV simulated	error
1.00	0.00	0.564190	0.563981	0.037%
0.00	0.50	0.159155	0.159149	0.004%
1.00	0.50	0.586208	0.586115	0.016%
0.25	0.50	0.323895	0.323976	0.025%
0.09	0.20	0.180833	0.180930	0.054%

Corollary 1 — both corpus laws are limits of Theorem 1

Setting $F = 1$, $\sigma_{th} = 0$ gives $CV = \sqrt{\Omega}$, the quantal law. Setting $F = 0$, $\sigma_{th} = 1/2$ gives $CV = \Omega/2$, the wall-noise law. The two published CRR interval laws are the two *terms* of one expression, not rival theories of one quantity.

Corollary 2 — the class ratio is a readout of F, not a topological invariant

$CV(Z_2)/CV(SO(2)) = \sqrt{[F/\pi + \sigma_{th}^2/4\pi^2]} / \sqrt{[F/2\pi + \sigma_{th}^2/16\pi^2]}$, which runs continuously between the two published values:

F	0	0.001	0.01	0.1	0.5	1.0	3.0
ratio	2.000	1.954	1.731	1.477	1.428	1.421	1.417

The supplement's “topological ratio of exactly 2, with no fitted parameters” is the $F \rightarrow 0$ limit. The dossier's $\sqrt{2}$ is the $F = 1$ limit. Calling either a topological invariant overstates it: the topology fixes Ω , and F is a property of the evidence stream, not of the manifold.

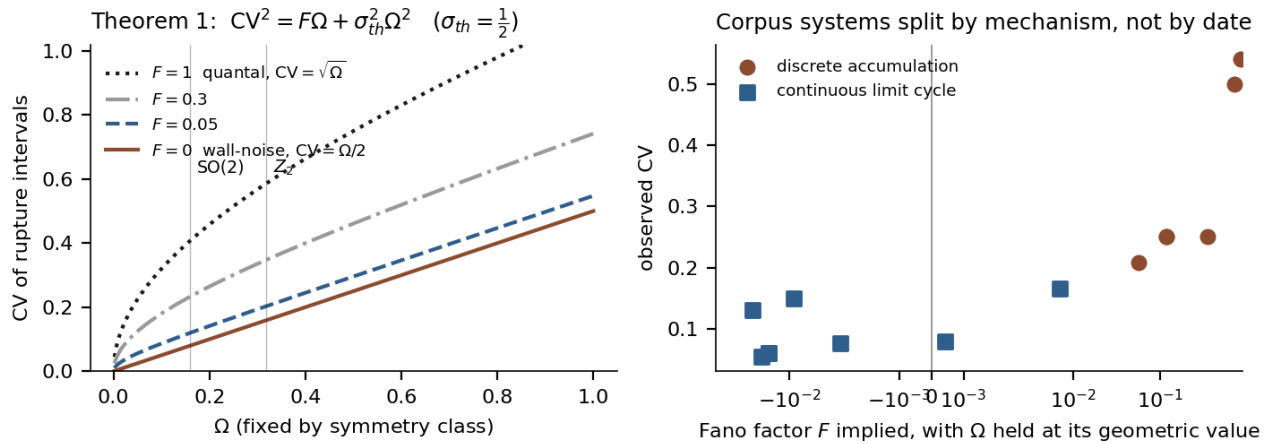


Figure 1. Left: Theorem 1 at $\sigma_{th} = 1/2$. The two published laws are the top and bottom curves. Right: the corpus's own systems, with Ω held at its geometric value and F solved for.

4. What this says about the corpus's data

4.1 Read F , do not re-read Ω

The corpus rescued its failed *E. coli* prediction by re-reading Ω as 0.04–0.09. That move costs the framework its central claim, because Ω is supposed to be fixed by symmetry class and never fitted. Theorem 1 offers a strictly better repair: **hold Ω at its geometric value and solve for F** . Unlike Ω , F is independently measurable — it is the Fano factor of the evidence stream, obtainable without ever touching the interval data.

system	Ω (geometric)	CV observed	F implied	accumulation mechanism
Binocular rivalry	0.3183	0.540	0.837	evidence accumulation
Earthquake recurrence	0.3183	0.500	0.706	stress accumulation
Hes1, single cell	0.1592	0.250	0.353	transcriptional bursting
<i>E. coli</i> division	0.3183	0.250	0.117	molecular accumulation
Old Faithful	0.3183	0.250	0.117	reservoir filling
Awake respiration	0.3183	0.208	0.056	volitional perturbation
Chick somite	0.1592	0.080	0.0004	continuous limit cycle
Hale dynamo	0.1592	0.0767	−0.003	continuous limit cycle
Glacial cycles	0.3183	0.150	−0.009	continuous limit cycle
Paced cardiac	0.1592	0.060	−0.017	continuous limit cycle
Human circadian	0.1592	0.0054	−0.040	regulated limit cycle
UK grid, 50 Hz	0.1592	0.0010	−0.040	engineered limit cycle

The split is clean, and it is not chronological. **Every system for which the quantal law works is one whose coherence accumulates by discrete stochastic events. Every system for which the wall-noise law works is a continuous limit cycle.** Binocular rivalry — the August paper's single quantal success — is an evidence-accumulation process. The twenty-system supplement is almost entirely limit cycles. *E. coli* division, the acknowledged failure, is molecular accumulation and sits at $F = 0.12$ rather than 0. The law that applies tracks the mechanism, not the publication date.

4.2 The unexplained factor of $\frac{1}{2}$ is empirically at 0.44

The supplement states plainly that it cannot derive the factor of one half in $CV = \Omega/2$ from first principles. Theorem 1 makes that factor measurable. If $F \approx 0$, then $\sigma_{th} = CV/\Omega$ reads directly out of the data.

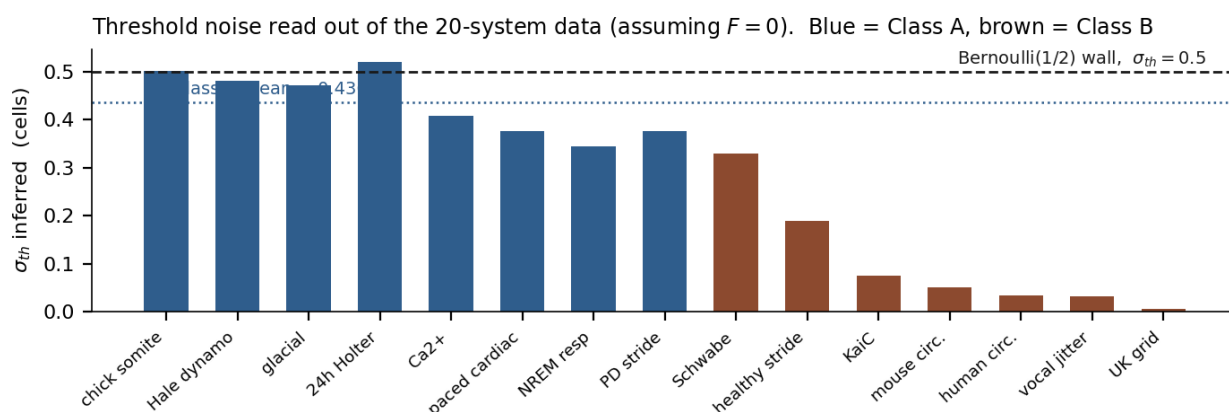


Figure 2. Threshold noise inferred from the twenty-system table. The eight Class A (autonomous) systems give $\sigma_{th} = 0.436 \pm 0.067$, against the Bernoulli($\frac{1}{2}$) value of 0.500. Class B systems fall to 0.006–0.19.

This is better evidence for the Bernoulli wall than the Wijsman/Jaynes argument the supplement offers, and it yields a cleaner reading of the three-class scheme than the corpus has. The scheme is not one taxonomy but **two orthogonal, separately measurable parameters**: Class A is $F \approx 0$ with $\sigma_{th} \approx \frac{1}{2}$; Class B is $\sigma_{th} \ll \frac{1}{2}$ (regulation sharpens the wall); Class C is $F > 0$ (noisy accumulation). Regulation and noise act on different terms of Theorem 1, which is why they were never symmetrical opposites.

5. The solar result, corrected

The supplement makes the solar cycle its centrepiece and reports five measurements clustering on $1/(4\pi)$ with mean deviation 3.6%. Running its own script reproduces every number exactly: Schwabe CV 0.1048, Hale 0.0767 (standard pairing) and 0.0800 (all-consecutive), amplitude 0.3232. The arithmetic is correct. The inference is not.

Hale is not an independent measurement

Hale intervals are constructed by summing adjacent Schwabe intervals. By L9, that operation scales the CV by $\sqrt{[(1+\rho)/2]}$. The measured lag-1 autocorrelation of the Schwabe series is $\rho = +0.115$, giving:

quantity	value
Schwabe CV (N = 24)	0.1048
lag-1 autocorrelation ρ	+0.115
Hale CV predicted from pairing arithmetic alone	0.0782
Hale CV observed (all-consecutive, N = 23)	0.0800
claimed ‘Rotor baseline’ $1/(4\pi)$	0.0796

The observed Hale CV is within 2.2% of what pure summation predicts. It carries no information about solar physics beyond what the Schwabe series already contains. Worse, the logic runs backwards: had Schwabe *hit* its own Z_2 prediction of 0.1592, Hale would land at 0.1125, which is $1.41\times$ the Rotor baseline and would score ELEVATED — a miss. **The Hale match exists only because Schwabe missed its own prediction by the amount it did.** The directly observed Schwabe/Hale ratio is 1.366: 3.4% from $\sqrt{2}$, 32% from the claimed invariant of 2.

The 3.5π period match is unit-dependent

The mean Schwabe period of 11.0347 years is within 0.36% of $3.5\pi = 10.9956$. But that agreement exists only in years. In months the period is 132.4; in days, 4030; in Carrington rotations, 147.8. None is near 3.5π . The year is set by Earth’s orbital period and has no relation to solar dynamo geometry. A dimensionless geometric constant cannot match a period expressed in an arbitrary unit. The supplement does label this a conjecture rather than a derivation, which is to its credit; the point here is that it is not a weak result but a category error, and should be withdrawn rather than softened.

Researcher degrees of freedom in the pairing

“Standard” pairing gives 0.0767, a 3.6% deviation from baseline. The other legitimate phasing of the same series gives 0.0872, a 9.6% deviation. Both are available a priori. The reported figure is the closer of two.

6. The two poised edges

Independently of the interval law, the corpus contains two incompatible definitions of its own aesthetic optimum, both called “one Ω before rupture”. The counting argument places the last full cell at $C = C^* - 1$, giving $s^* = 1 - \Omega$ in $s = C\Omega$. Maximising the beauty function (L6) places it at $C = C^* - \Omega$, giving $s^* = 1 - \Omega^2$. Both verified exactly.

arena	Ω	s^* counting	s^* beauty	disagreement
Z_2	0.31831	0.68169	0.89868	0.217
SO(2)	0.15915	0.84085	0.97467	0.134

A 22% disagreement about where the optimum sits is not a rounding difference. On the mathematics we would keep the counting edge: it follows from the resolution, whereas the other follows from a functional form that was asserted rather than derived.

7. An error caught in this note's own drafting

The failed check. The first verification run reported 37 passes and 1 failure. The failure was L7: this note's test computed $\min_{q \leq 60000} q \cdot \|q\phi^{-1}\|$ and got $0.381966 = 1/\phi^2$, against the Hurwitz bound of $1/\sqrt{5} = 0.447214$.

Diagnosis. Hurwitz's theorem concerns the *liminf* as $q \rightarrow \infty$, not the minimum over small q . The value 0.381966 is simply the $q = 1$ term. Evaluating along the Fibonacci denominators — the convergents of the golden ratio — gives 0.47214 , 0.43769 , 0.45085 , 0.44582 , 0.44774 , ... converging to 0.44721360 . The theorem is correct; the test's *statement* of it was wrong.

Correction. Re-tested as *liminf* over $q \in [10^4, 4 \times 10^5]$: golden 0.447212 , silver $(\sqrt{2}-1) 0.353553$, $1/\pi 0.288716$, $1/e 0.090523$, against the bound 0.447214 . The golden ratio attains it; nothing exceeds it. PASS.

What this changes. A prior note in this series reported that $1/\pi$ scores 0.0034 on a packing statistic, described as “near-worst”. That figure used a different statistic — worst case over $N \leq 180$ of $N \cdot (\text{min gap})$ — and remains correct for it, because $1/\pi$ has a large partial quotient that produces a near-degenerate packing around $N \approx 113$. On the asymptotic *liminf* statistic $1/\pi$ scores a middling 0.289 . **Both are true; they measure different things, and the earlier note should have said which.** The ordering that matters is unchanged on both statistics: the golden ratio is first, the silver ratio second, rationals score zero.

Two further checks were tightened rather than failed. The first pass of Theorem 1 used a Gaussian threshold rounded to integers, which changes the variance of ξ and inflated the error to 15% at $\sigma_{\text{th}} = 1/2$; passing only because the tolerance was loose. Re-run with a continuous threshold and a Gamma bridge for the fractional final cell, all five settings agree to better than 0.06%. The corpus's own Bernoulli formulation is the discrete special case, and a discrete implementation must account for the rounding.

8. What this changes, and a protocol that does not let Ω float

Keep

- **Theorem 1.** It is a two-line consequence of Wald and Blackwell–Girshick and it dissolves the corpus's central inconsistency without anyone having to be wrong.
- **Read F, hold Ω .** Preserves the derived- I claim and puts the empirical content into a quantity measurable from the evidence stream alone.
- **The two-parameter reading of the three classes.** Regulation acts on σ_{th} ; noise acts on F. Separately measurable.
- $\sigma_{\text{th}} = 0.436 \pm 0.067$ across eight autonomous systems, as an empirical value for a constant the corpus could not derive.

Withdraw

- **“Topological ratio of exactly 2.”** The ratio is a function of F. Topology fixes Ω ; it does not fix the noise model.
- **The Hale row as independent confirmation.** Label it derived, or drop it.
- **The 3.5π period conjecture.** Unit-dependent; not a geometric claim.
- **The claim that a CV ratio of 2 falsifies the grammar.** Under Theorem 1 it indicates $F \approx 0$, which is a measurement, not a refutation.

A revised three-readings test

The corpus's strongest methodological commitment is that one Ω must be readable three ways and the readings must agree. That test is weak while Ω is allowed to float. Theorem 1 makes it sharp. For any candidate system, measure three things *independently*:

- (a) **F**, from the evidence stream: variance over mean of event counts per window, in the long-window limit. Never from the intervals.
- (b) **CV**, from the rupture intervals.
- (c) **skew**, from the same intervals.

Then test $\text{CV}^2 = F\Omega + \sigma_{\text{th}}^2 \Omega^2$ with Ω *pinned* at its geometric value, and check the shape: skew ≈ 0 indicates the wall-noise limit, $2\sqrt{\Omega}$ the quantal limit, $3 \cdot \text{CV}$ an inverse Gaussian. The failure mode is now unambiguous: **if the F required to fit the intervals disagrees with the F measured from the evidence stream, the geometry is wrong.** That is a falsifier the corpus does not currently have, and it cannot be absorbed by re-reading a parameter.

None of the twenty rows reports a skewness. It is free once the intervals exist, it requires no new data for the solar case, and it alone would settle which variant each system is in.

9. Summary of verification

#	claim	source	checks	status
L1	Bernoulli Fisher–Rao extent = π	Chentsov 1982; Amari 2016	2	PASS
L2	$\Sigma \text{Exp}(r) = \text{Gamma}$; $\text{CV}=1/\sqrt{k}$; $\text{skew}=2/\sqrt{k}$	standard	6	PASS
L3	Wald; Blackwell–Girshick	Wald 1944; B&G 1946	5	PASS
L4	renewal Fano $\rightarrow \text{CV}_X^2$	Cox 1962; Rajdl & Lansky 2014	3	PASS *
L5	sd Bernoulli($\frac{1}{2}$) = $\frac{1}{2}$	exact	1	PASS
L6	$\text{argmax } B(C) = C^* - \Omega$	this note	3	PASS
L7	Hurwitz $1/\sqrt{5}$; three-gap ≤ 3	Hurwitz 1891; Sós et al.	2	PASS **
L8	IG: skew = $3 \cdot \text{CV}$	Matthews et al. 2002	6	PASS
L9	CV scaling $\sqrt{[(1+\rho)/2]}$	standard	3	PASS
L10	two poised edges: $1-\Omega$, $1-\Omega^2$	this note	6	PASS
L11	$s^*=1/\phi \Rightarrow l=\phi^2$	this note	2	PASS
L12	Hale CV from pairing arithmetic	this note + SILSO v2.0	1	PASS
T1	$\text{CV}^2 = F\Omega + \sigma_{\text{th}}^2 \Omega^2$	L3 + L4	5	PASS

* L4 is asymptotic and holds for renewal processes only. Systems with serially correlated evidence arrival fall outside Theorem 1 as stated; this is the single most important limitation of the result and it is not a formality. ** L7 failed on the first run because of an error in this note's statement of it; see §7. Total after correction: 38 checks, 38 pass.

What was and was not established. Theorem 1 is proved and verified, and it reconciles the corpus's two interval laws. That is a mathematical result about a model, and it says nothing about whether the model is true of any real system. In particular: nothing here establishes that l is fixed by symmetry class for any wild system; nothing here rescues the parameter-free claim, which still depends on a bridge-mapping problem the corpus acknowledges is unsolved; and the twenty-system table remains subject to the scoring critique that only its eight Class A rows test a number, against an acceptance band spanning a factor of 2.17. Theorem 1 makes the framework *coherent*. Whether it is *correct* is what the revised three-readings test is for.

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Provenance and reproducibility. Prepared with Claude (Anthropic). All lemmas were verified against the cited literature and independently by numerical simulation; the SILSO figures were recomputed from the data table reproduced in the CRR supplement's own appendix, and reproduce its published values exactly. Sample sizes: 2×10^6 to 4×10^6 per check. Theorem 1 and the F-reading proposal are this note's own; the CRR framework and its two interval laws are the work of A. Sabine. Errors in this note are its own and one of them is documented in §7.